Functional Requirement Document

Olist- E-Commerce Business Performance Analysis

Document Type :	Functional Requirements Document (FRD)	Status :	Final
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Project Name :	Olist E- Commerce Analytics	Tool Stack :	SQL, Power BI, DAX

System Overview

- Data Preparation (MySQL 8.0): Raw CSV files are ingested into a relational MySQL database. Data cleaning, type conversion, indexing, and deduplication are performed at this layer before any analysis begins.
- SQL Analysis: 9 advanced SQL queries are executed against the cleaned database to extract business insights. Queries use window functions, CTEs, multi-table JOINs, and aggregation to answer specific business questions.
- Power BI Dashboard: MySQL is connected directly to Power BI Desktop. A star-schema data model is built, 12+ DAX measures are engineered, and a 7-page interactive dashboard is constructed for stakeholder consumption.

Functional Requirements

FR-01 — The system shall ingest all 7 source CSV files into a MySQL relational database with correct data types and without data loss.

FR-02 — The system shall clean all DateTime columns by converting empty string values to NULL to enable delivery time calculations.

FR-03 — The system shall deduplicate the category translation table and resolve hidden character encoding issues before any category-level analysis.

FR-04 — The system shall create indexes on all foreign key columns to ensure query performance on 100,000+ row joins.

FR-05 — The system shall produce a monthly revenue trend report showing total orders and total revenue grouped by year and month for all delivered orders.

FR-06 — The system shall rank all product categories by total revenue, order volume, and average product price.

FR-07 — The system shall segment all customers into spend tiers using a percentile-based scoring method and calculate revenue contribution per tier.

FR-08 — The system shall apply RFM scoring to all customers using Recency, Frequency, and Monetary dimensions and assign each customer a segment label.

FR-09 — The system shall calculate average delivery days and late delivery rate for each of the 27 Brazilian states.

FR-10 — The system shall analyse seller revenue distribution and calculate the revenue share % controlled by top seller groups.

FR-11 — The system shall deliver a multi-page interactive dashboard with dynamic filtering by state, year, and product category.

FR-12 — The system shall include a recommendations page that translates analytical findings into a prioritised business action plan.

User Stories

US-01

As a business manager, I want to see monthly revenue trends so that I can identify whether the business is growing, declining, or seasonal.

US-02

As a marketing head, I want product categories ranked by revenue so that I can direct marketing budget toward the highest-performing categories.

US-03

As a regional manager, I want order volume and revenue broken down by state so that I can identify which regions to prioritise for expansion.

US-04

As a seller management team, I want sellers ranked by revenue and order volume so that I can identify top performers and flag underperformers for review.

US-05

As a finance team, I want payment method breakdown and installment patterns so that I can understand how customers prefer to pay and optimise checkout accordingly.

US-06

As a CRM team, I want customers segmented by lifetime spend so that I can focus retention investment on the most valuable customer group.

US-07

As a CRM team, I want RFM-based customer segmentation so that I can identify customers at risk of churning before they are lost.

US-08

As a logistics manager, I want average delivery days and late delivery rates by state so that I can identify regional bottlenecks and take corrective action.

US-09

As a business strategist, I want to understand how revenue is distributed across sellers so that I can assess whether the platform has dangerous dependency on a few sellers.

US-10

As a C-suite stakeholder, I want a self-service dashboard with dynamic filtering so that I can answer business questions without waiting for manual reports.

US-11

As a C-suite stakeholder, I want a prioritised recommendations page so that I can take immediate data-backed action on the most critical business problems.

SQL QUERY SPECIFICATIONS

Each SQL query must return the defined output columns, filter correctly on delivered orders only , and produce reproducible results :

FR ID	Query Name
FR.1	Monthly Revenue Trend
FR.2	Top Revenue Categories
FR.3	Top States by Revenue
FR.4	Best and Worst Sellers
FR.5	Payment Method Analysis
FR.6	Customer Spend Segmentation
FR.7	RFM Segmentation
FR.8	Delivery Time by State
FR.9	Seller Concentration Risk

POWER BI DATA MODEL SPECIFICATIONS

Star Schema Design The Power BI data model must implement a star schema with order_items as the central fact table. All dimension tables connect to the fact table via single-direction relationships with Many-to-One cardinality.

FR ID	From Table	From Column	To Table	To Column	Cardinality	Direction
FR-1	order_items	order_id	orders	order_id	Many-to-One	Single
FR-2	order_items	product_id	products	product_id	Many-to-One	Single
FR-3	order_items	seller_id	sellers	seller_id	Many-to-One	Single
FR-4	orders	customer_id	customers	customer_id	Many-to-One	Single
FR-5	order_payments	order_id	orders	order_id	Many-to-One	Single
FR-6	products	product_category_name	category_translation	product_category_name	Many-to-One	Single
FR-7	Calendar	Date	orders	order_purchase_timestamp	One-to-Many	Single

Calendar Table: A calculated Calendar table must be created using DAX CALENDAR() function spanning 01-Jan-2016 to 31-Dec-2018, marked as Date Table, with the following derived columns: Year (INT), Month (INT), Month Name (TEXT), Quarter (TEXT: Q1–Q4), Year-Month (TEXT: YYYY-MM). Time intelligence DAX measures depend on this table.

DASHBOARD PAGE SPECIFICATIONS

Page 1 — Executive Overview

Business Question: What is the overall business performance at a glance?

Required Visuals:

- → KPI Cards (×4): Total Revenue, Total Orders, Total Customers, Avg Order Value
- → Line Chart: Total Orders by Month — X-axis: Month, Y-axis: Total Orders
- → Filled Map: Total Orders by State — Location: customer_state, Saturation: Total Orders

- → Donut Chart: Orders by order_status — Legend: order_status, Values: Total Orders

Slicers: State (Dropdown) | Year (Dropdown)

Page 2 — Sales & Revenue

Business Question: Where is revenue coming from and how is it growing year-over-year?

Required Visuals:

- → KPI Cards (×3): Total Freight Revenue, Total Revenue, Product Revenue
- → Bar/Column Chart: Revenue Over Years and Months
- → Matrix Table: Revenue Contribution & Growth — Rows: category_name, Columns: Revenue, % of Revenue, Growth %
- → Donut Chart: Mode of Payment Distribution — payment_type vs payment_value
- → Clustered Bar Chart: Revenue Growth — Total Revenue vs Previous Year Revenue by State

Slicers: Year (Dropdown) | Category (Dropdown) | State (Dropdown)

Filters: Delivered orders only

Page 3 — Customer Intelligence

Business Question: Who are our customers, where are they, and are we retaining them?

Required Visuals:

- → KPI Cards (×3): Avg Revenue Per Customer, Total Customers, Repeat Customers
- → Donut Chart: One-Time vs Repeat Customers — Customer segment breakdown
- → Line Chart: Total Customers Trend — monthly trend Jan 2017 to Dec 2017
- → Bar Chart: Total Revenue by customer_state — top 10 states
- → Bar Chart: Top 10 Cities by Customers

Slicers: State (Dropdown) | Year (Dropdown) | Category (Dropdown)

Filters: Delivered orders only

Page 4 — Seller & Supply Chain Performance

Business Question: Which sellers are performing and is there revenue concentration risk?

Required Visuals:

- → KPI Cards (×3): Total Sellers, Revenue Per Seller, Top Seller Revenue %
- → Bar Chart: Top 20 Sellers by Revenue — seller_id vs Total Revenue
- → Bar Chart: Seller Revenue Distribution — revenue bucket bands (R$1K, R$1K-5K, etc.)
- → Matrix Table: Regional Seller Metrics — seller_state, total sellers, cities, orders, revenue
- → Filled Map: Revenue by seller_state

Slicers: State (Dropdown) | Year (Dropdown) | Category (Dropdown)

Filters: Delivered orders only

Page 5 — Product & Category Insights

Business Question: Which categories drive revenue vs volume vs average product price?

Required Visuals:

- → KPI Cards (×3): Total Products, Total Categories, Top Category + Revenue Share %
- → Bar Chart: Revenue by Category — all 69 categories ranked by revenue
- → Bar Chart: Orders by Category — all categories ranked by order volume
- → Bar Chart: Categories by Avg Product Price — identifies premium vs budget categories

Slicers: Year (Dropdown) | State (Dropdown) | Category (Dropdown)

Page 6 — Logistics & Delivery

Business Question: Where are deliveries failing and which states have the worst performance?

Required Visuals:

- → KPI Cards (×4): Avg Delivery Days, Avg Estimated Days, States with <5% Late Deliveries, On-Time Delivery %
- → Scatter Chart: Revenue vs Freight Cost % by State
- → Heat Matrix: Monthly Late Order % by State — conditional formatting (red=high, green=low)
- → Line Chart: On-Time vs Late Delivery % trend by Month (toggle buttons)
- → Dual Bar Chart: On-Time vs Late Delivery % by State — bottom 6 worst performers

Slicers: Year (Dropdown) | States (Dropdown) | Category (Dropdown)

Page 7 — Recommendations

Business Question: What should the business prioritise to improve performance?

Required Visuals:

- → KPI Cards (×4): Top 5 Category Revenue %, Top Seller Revenue %, On-Time Delivery %, Repeat Customers
- → Text Box: Revenue Growth finding with bold key metrics and actionable recommendation
- → Flowchart (Shapes): Execution Roadmap — Immediate → Next → Scale with action descriptions
- → Button Row: Recommendation For — Customer Retention | Logistics Excellence | Revenue Growth | Seller Management

Slicers: No slicers — static summary page

12. ASSUMPTIONS & CONSTRAINTS

Assumptions

• Revenue is defined as price + freight_value for all delivered orders only. • customer_unique_id is used (not customer_id) to correctly identify distinct individuals across multiple orders.

• The reference date for RFM recency calculation is 2018-08-31 — the last date in the dataset.

• order_reviews table is excluded due to irreparable special character encoding issues during CSV import.

• Geolocation table (1M rows) is loaded directly in Power BI as CSV to bypass MySQL import performance issues.

Constraints

• Dataset covers September 2016 to August 2018 only — no data beyond this window.

• Power BI Desktop (free version) used — no Publish to Web or Row-Level Security features.

• MySQL Community Server 8.0 — no stored procedures or triggers used.

• No real-time data refresh — static dataset loaded once.